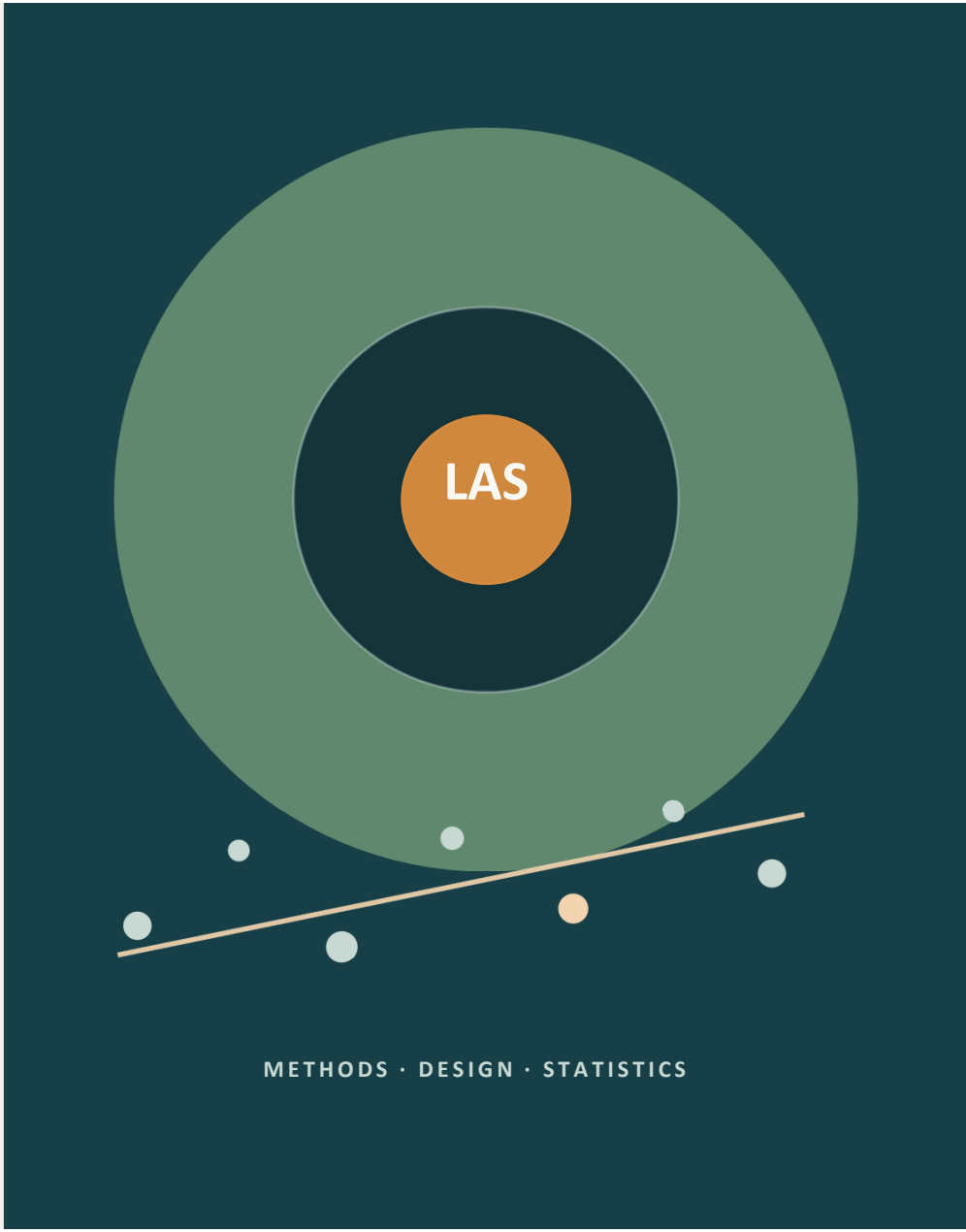
PRINCIPLES OF LABORATORY ANIMAL SCIENCE

Methodological and statistical aspects

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UMCG · Department of Epidemiology



Agenda

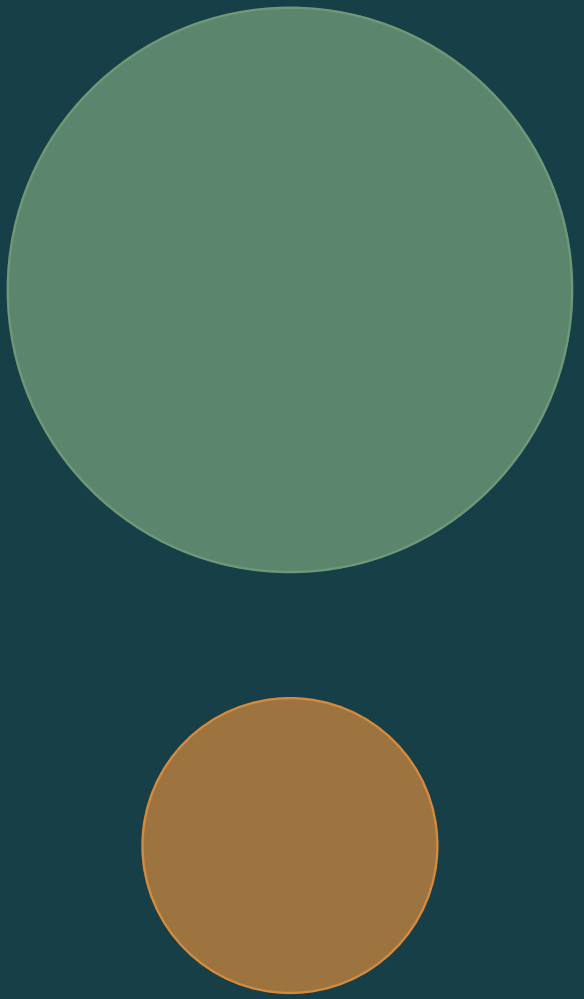
1 **Statistical inference in animal experiments**
Treatment-effect estimation, precision, confidence intervals and hypothesis testing.

2 **Sample size calculation**
Type II error, power and planning the number of animals.

3 **Handling nuisance variables**
Animal characteristics, experimental context, stratified randomization and blocking.

PART I

Statistical inference in animal experiments



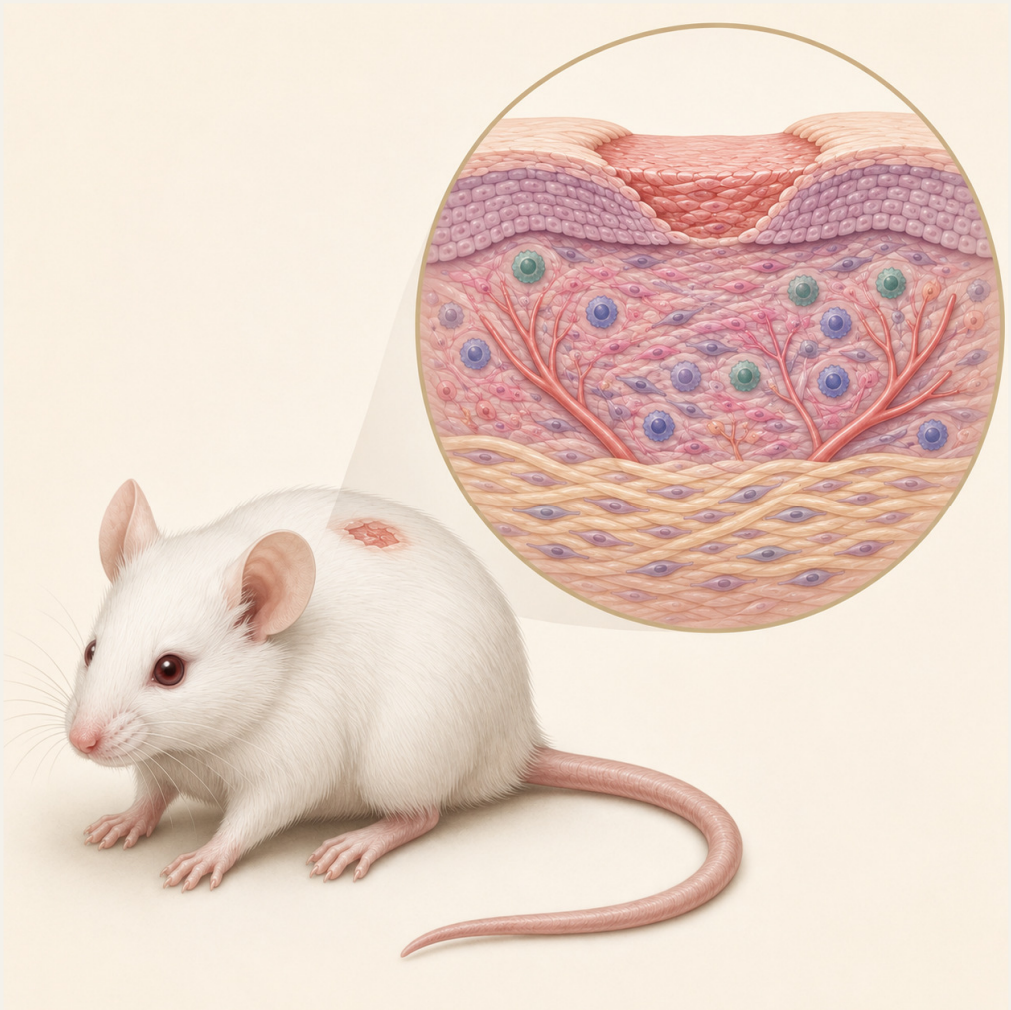
Gel X and wound healing: scientific problem and question

SCIENTIFIC PROBLEM

- Wound healing requires a coordinated process of tissue repair.
- Some wounds heal slowly or incompletely, prolonging tissue damage.
- Gel X is proposed to support repair, but its effect must be established.

SCIENTIFIC QUESTION

What is the effect of Gel X on wound healing in adult mice?



Gel X versus vehicle: a randomized controlled experiment

EXPERIMENTAL DESIGN

- Scientific target population: adult mice
- Study animals: 24 healthy C57BL/6J mice aged 10–12 weeks
- Random allocation: 12 mice to topical Gel X and 12 to vehicle
- Vehicle control: the same gel base without the active ingredient
- Outcome: percentage wound closure after seven days

STUDY OBJECTIVE

What is the effect of Gel X versus vehicle on day-7 wound closure?

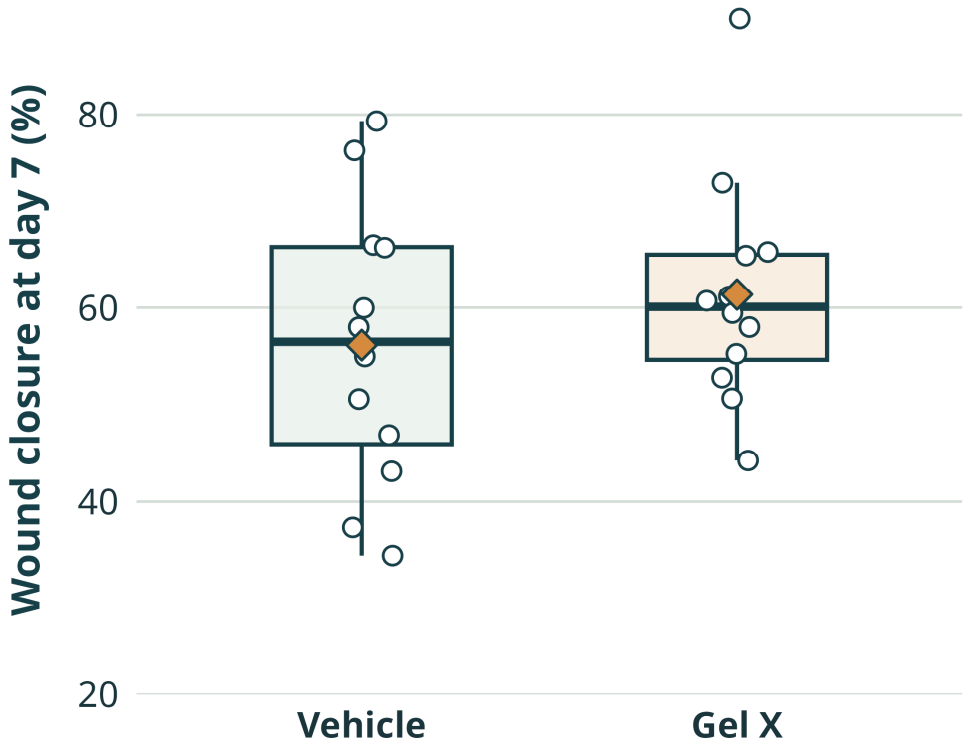


Descriptive analysis of the Gel X study

Observed treatment groups

CHARACTERISTIC	VEHICLE (n = 12)	GEL X (n = 12)
Female, n (%)	8 (66.7)	4 (33.3)
Body weight, g	21.4 (2.2)	23.2 (2.3)
Initial wound area, mm²	28.6 (1.8)	27.9 (1.6)

Mean (SD) unless otherwise stated



Vehicle mean 56.1% | Gel X mean 61.4% | Observed difference +5.2 percentage points

Estimating the population treatment effect

1 **What do we want to estimate?**
The difference in population mean wound closure under Gel X versus vehicle.
$\Delta = \mu_{\text{Gel X}} - \mu_{\text{vehicle}}$

2 **How do we estimate it?**
Use the difference between the observed group means: +5.2 percentage points.

3 **What contributes to this observed difference?**
Observed group difference = true treatment effect
+ contributions from other observed and unobserved factors that differ between the groups
Examples: sex, body weight and initial wound severity.

4 **When is this estimator unbiased?**
With random allocation, these other contributions average to zero across repeated experiments. The estimates are therefore centred on the true treatment effect.

How much would this estimate vary across repeated experiments?

How precise is our treatment-effect estimate?

Live simulation: population means 56% (vehicle) and 66% (Gel X) → Δ = 10 percentage points

EACH EXPERIMENT GIVES AN ESTIMATE

Repeat with 24 mice: randomly allocate 12 to each treatment.



THE SPREAD OF ESTIMATES DESCRIBES PRECISION



For an approximately Normal sampling distribution, roughly 68% of estimates fall within one SE of the true treatment effect.

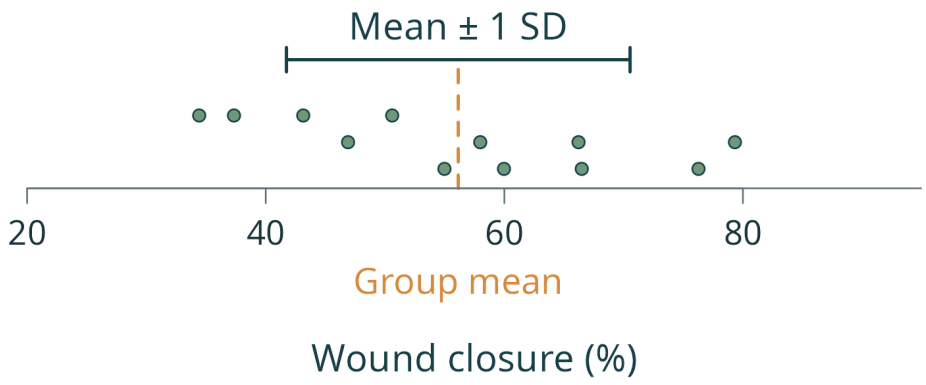
The standard error is a measure of how much the treatment-effect estimate varies across repeated experiments.

Smaller SE
Greater precision

Standard deviation and standard error: two different kinds of spread

Individual animals vary within an experiment. Treatment-effect estimates vary between experiments.

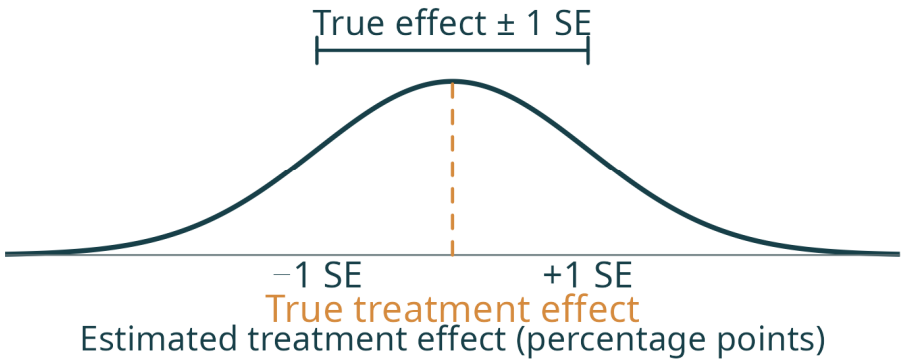
VARIATION BETWEEN ANIMALS — SD



SD describes how widely individual outcomes are scattered around their group mean.

More variation between animals → larger SE (same sample size)

VARIATION BETWEEN EXPERIMENTS — SE



SE estimates the spread of treatment-effect estimates across experiments.

More animals per group → smaller SE (same outcome variability)

More animals can improve precision without making individual animals less variable.

Confidence interval: which treatment effects are compatible with the data?

FROM ESTIMATE TO INTERVAL

1 **Point estimate**

Δ̂ = +5.2 percentage points

The observed difference between the group means.

2 **Standard error**

SE(Δ̂) = 5.4 percentage points

The estimated sampling variability of Δ̂.

3 **95% confidence interval**

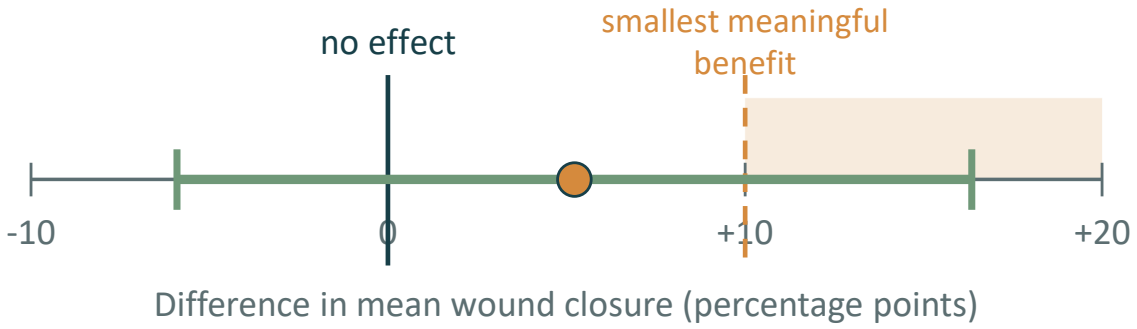
Δ̂ ± k × SE(Δ̂)

For a 95% CI, k ≈ 2. Its exact value depends on the assumptions underlying the statistical analysis, such as those made about the population model.

GEL X CASE

+5.2 percentage points

95% CI −5.9 to +16.3



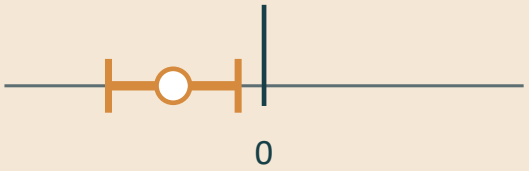
Under the model, treatment effects from −5.9 to +16.3 percentage points remain reasonably compatible with these data.

What can we conclude from the confidence interval?

Gel X – vehicle: negative favours vehicle · zero means no effect · positive favours Gel X

CI ENTIRELY BELOW 0

Example: 95% CI −12 to −2

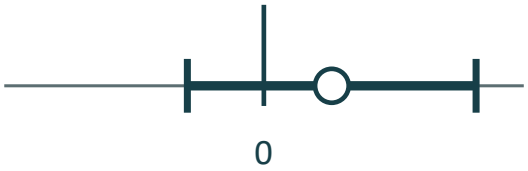


Inferiority established

Evidence that vehicle produces greater mean wound closure.

CI INCLUDES 0

Gel X: 95% CI −5.9 to +16.3

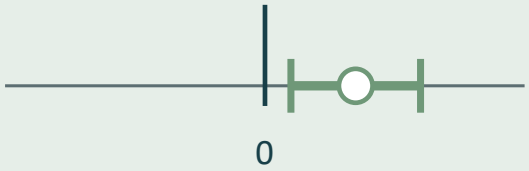


Neither established

Neither superiority nor inferiority of Gel X is established.

CI ENTIRELY ABOVE 0

Example: 95% CI +2 to +12



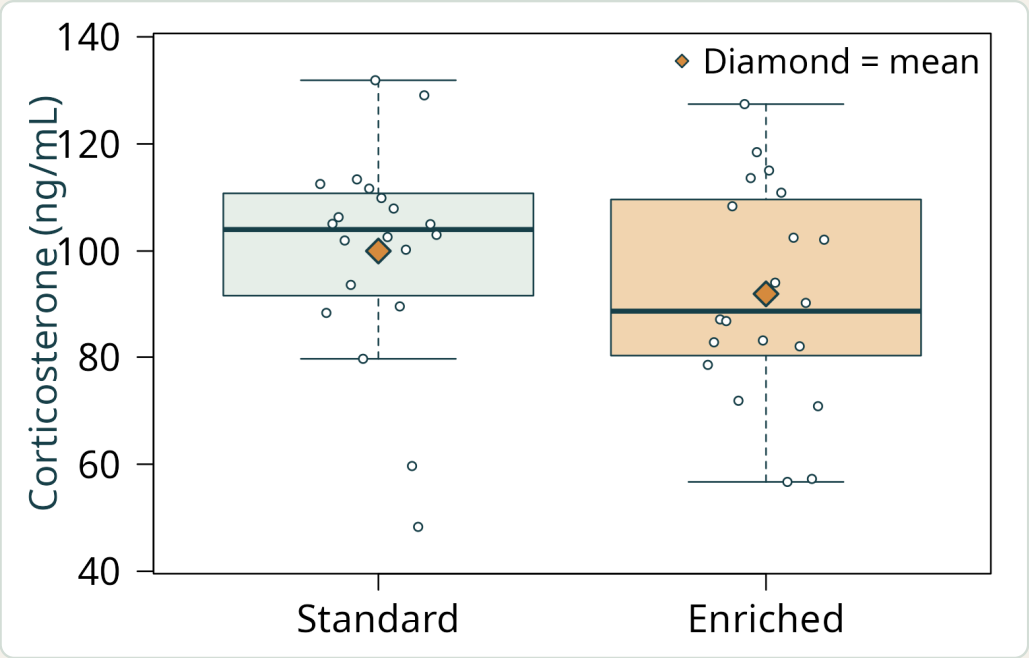
Superiority established

Evidence that Gel X produces greater mean wound closure.

Our interval includes negative and positive effects. Neither superiority nor inferiority of Gel X is established.

Exercise: does an enriched recovery environment reduce stress?

Forty individually housed mice were randomly allocated to standard or enriched recovery housing (20 per group). After a standardized procedure, plasma corticosterone was measured as an indicator of stress.



RESULTS (ng/mL)

Standard: mean 100 · SD 20

Enriched: mean 92 · SD 20

Difference (enriched − standard): −8.0

SE of the difference: 6.3

Simulated teaching data. Lower corticosterone indicates less stress in this example.

Exercise: randomization and sample size

Standard versus enriched recovery housing · 20 mice per group

1. What is the role of randomization in this experiment?

2. Suppose we repeated the experiment with four times as many animals from the same population.

    Would you expect the SDs to increase, decrease or remain similar?
    Would you expect the SE to increase, decrease or remain similar?

3. Would increasing the sample size necessarily produce a larger estimated treatment effect?

Exercise: confidence interval and conclusion

Enriched – standard: –8.0 ng/mL · SE: 6.3 ng/mL · Lower values indicate less stress

1. Calculate an approximate 95% confidence interval using:

   estimated difference ± 2 × SE

2. Does this interval establish superiority of enriched housing, inferiority of enriched housing, or neither?

Hypothesis testing: assessing a claim about the treatment effect

A hypothesis test asks whether the evidence is sufficient to reject a specified claim.

SPECIFY THE CLAIM

Null hypothesis (H_0): the true treatment effect is zero.

Gel X and vehicle have the same population mean wound closure.

The two-sided alternative is that their population means differ.

COMPARE THE DIFFERENCE WITH ITS SE

The t-statistic measures how far our estimate lies from the claimed value, relative to its uncertainty.

t = (estimated effect − 0) / SE

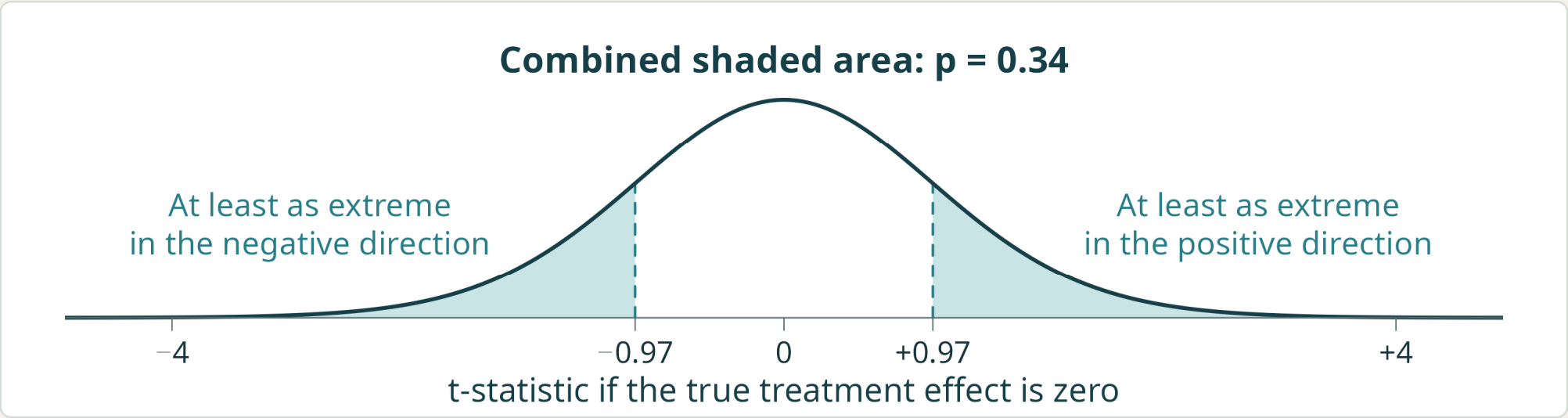
For Gel X: t = 5.2 / 5.4 ≈ 0.97

Our observed difference is approximately one standard error away from zero.

How unusual is our result under the null hypothesis?

Assume the true treatment effect is zero. How often would a t-statistic at least this far from zero occur?

Independent-samples t-test · observed t = 0.97 · include equally extreme results in either direction



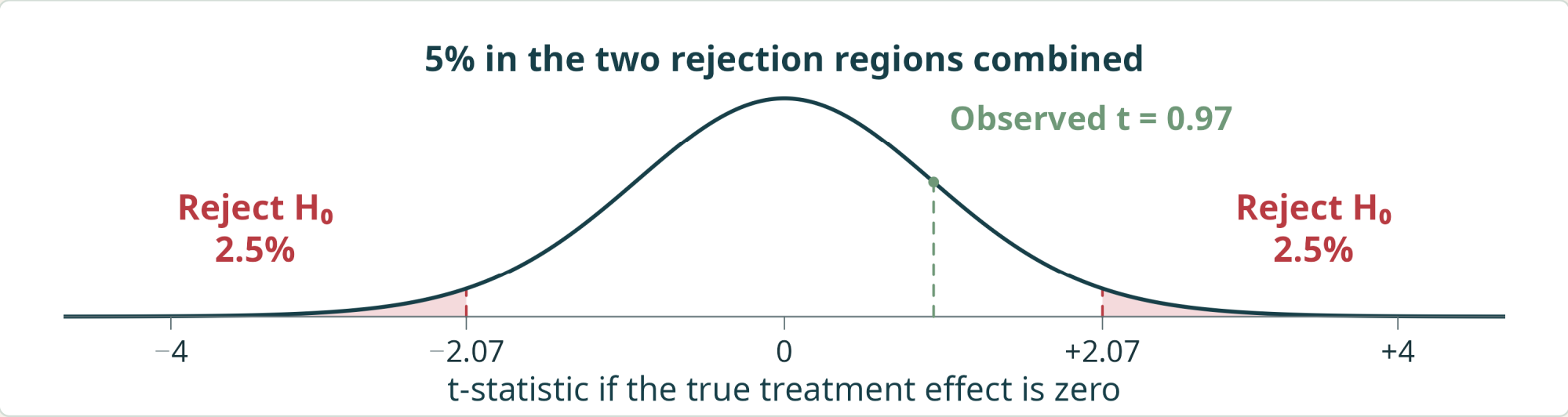
Under no treatment effect, about 34% of experiments would produce a t-statistic at least this extreme. Our result is not particularly unusual under H_0.

The p-value is the probability, under H_0, of a result at least as extreme as the one observed.

From a p-value to a decision

Before examining the results, specify $\alpha = 0.05$. Reject H_0 when $p < \alpha$.

This rule rejects H_0 in the two red tails: together, 5% of the null distribution.



A Type I error means rejecting H_0 when it is true. At the 5% significance level, this happens in 5% of repeated experiments with no treatment effect.

Gel X: $p = 0.34 > 0.05$. We do not reject H_0; the observed t lies outside the rejection regions.

The hypothesis test and confidence interval give the same decision

HYPOTHESIS TEST

Independent-samples t-test: p = 0.34

The p-value exceeds 0.05.

We do not reject the null hypothesis of zero treatment effect.

95% CONFIDENCE INTERVAL

Gel X: −5.9 to +16.3 percentage points

The interval includes zero.

Neither superiority nor inferiority is established.

For the corresponding two-sided test at the 5% level:
reject zero as the treatment effect exactly when zero lies outside the 95% confidence interval.

The interval also shows which effect sizes remain compatible with the data.

A test decision is not the full scientific conclusion

Both confidence intervals include 0, but they support different conclusions about a meaningful benefit.

Meaningful benefit is not supported

+1.0 pp (95% CI −3.0 to +5.0)

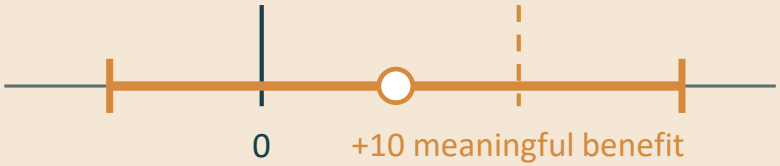


The interval excludes benefits of 10 percentage points or more.

The data do not support a benefit of 10 percentage points or more.

Meaningful benefit remains possible

+5.2 pp (95% CI −5.9 to +16.3)



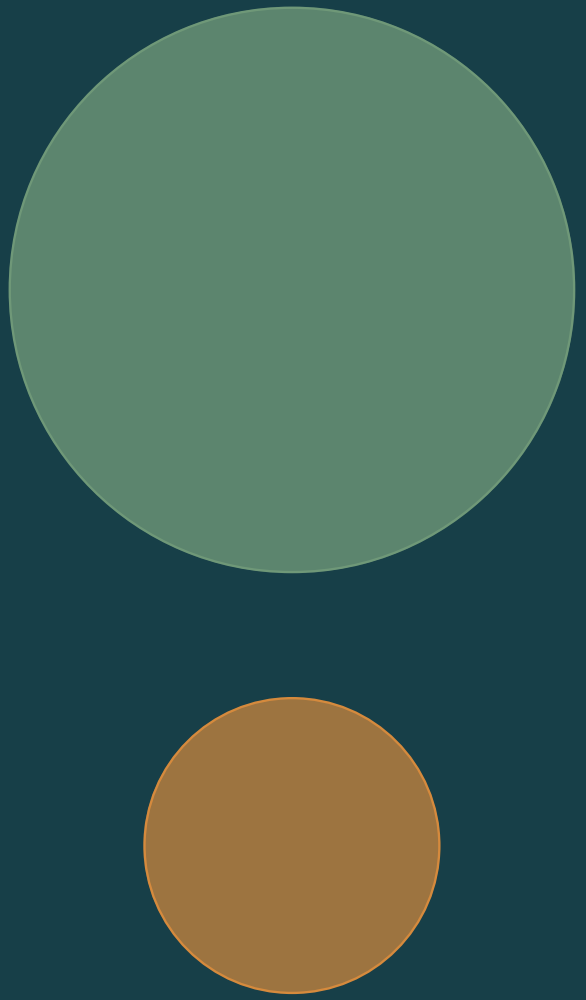
Compatible with no effect and with a benefit of at least 10 percentage points.

The study is too imprecise to determine whether Gel X provides a meaningful benefit.

Our Gel X experiment remains inconclusive about a meaningful benefit. How many animals would we need to plan for?

PART II

Sample size calculation



From Type I error to Type II error and power

We have considered the risk of concluding that an effect exists when it does not. For sample-size planning, we also need to consider the risk of missing an effect that does exist.

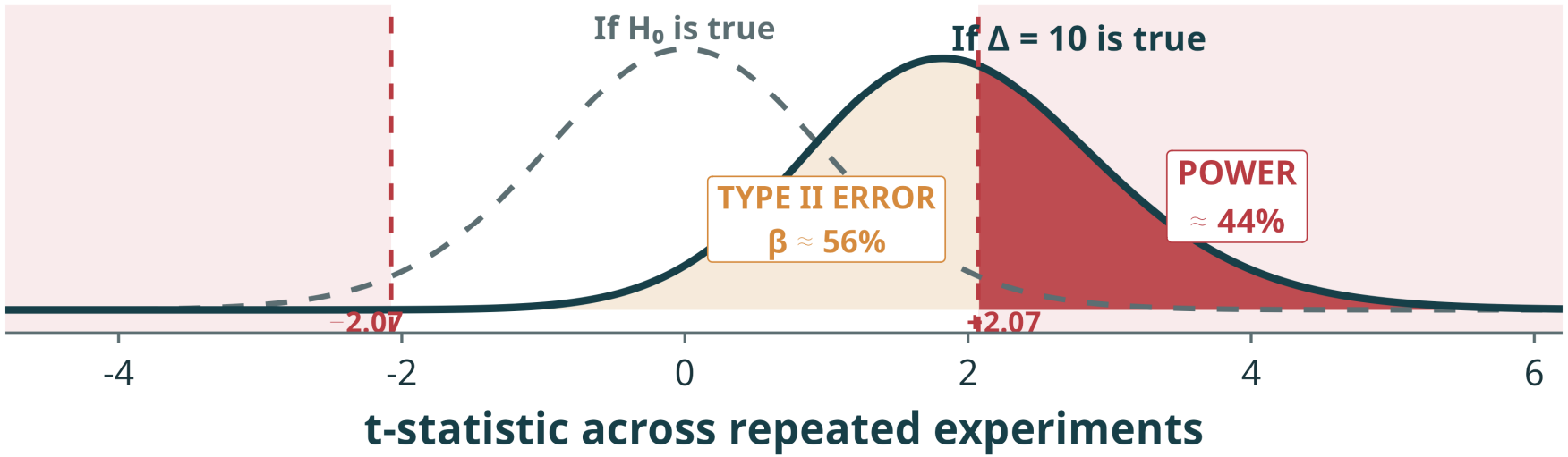
REALITY	DO NOT REJECT H_0	REJECT H_0
H_0 is true **$\Delta = 0$**	**CORRECT DECISION** Do not reject H_0	**TYPE I ERROR** Reject H_0 when the true effect is zero. Probability: α
Specified effect is true **$\Delta = \delta$**	**TYPE II ERROR** Do not reject H_0 when the specified effect δ is true. Probability: β	**POWER** Correctly reject H_0 when the specified effect δ is true. Probability: $1 - \beta$

Power is the probability of detecting a specified treatment effect, if that effect is truly present.

When Δ = 10 is true, the alternative distribution determines power

REALITY	TYPE II ERROR	POWER
Specified effect is true: Δ = 10	Do not reject H_0	Correctly reject H_0

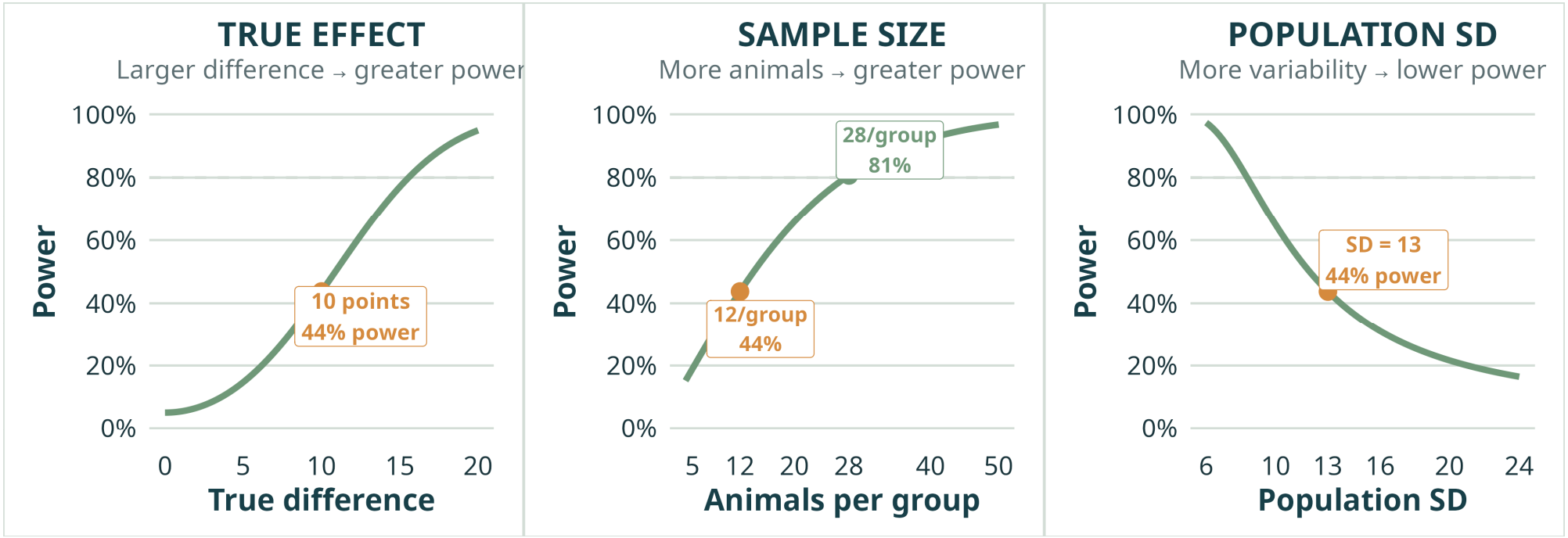
True effect = 10 percentage points · population SD = 13 points · N = 24 (12 per group) · two-sided α = 0.05



With these assumptions, approximately 44% of repeated experiments reject H_0: the power is 44%.

Power depends jointly on the true effect, sample size and population variability

Two-sided α = 0.05 and equal allocation are held constant; one factor changes in each panel.



Sample-size planning fixes the effect, SD and desired power, then solves for the required number of animals.

A Normal approximation gives the required number of animals per group

For two independent treatment groups with equal allocation and a common population SD:

$$n \approx \frac{2\sigma^2 \left(z_{1-\alpha/2} + z_{1-\beta}\right)^2}{\delta^2}$$

n = animals per group
σ = anticipated within-group SD
δ = treatment difference to detect

α = two-sided significance level
1 − β = desired power

z_p is the value below which proportion p of the standard Normal distribution lies.

Always round upward. An exact t-test calculation can give a slightly larger sample size.

Standard Normal percentiles for sample-size planning

p	z_p
0.80	0.842
0.85	1.036
0.90	1.282
0.95	1.645
0.975	1.960
0.99	2.326
0.995	2.576

TWO-SIDED SIGNIFICANCE LEVEL

Look up $p = 1 - \alpha/2$.
For $\alpha = 0.05$: $1 - \alpha/2 = 0.975$,
so $z_{0.975} = 1.960$.

DESIRED POWER

Look up $p = 1 - \beta$.
For 90% power: $1 - \beta = 0.90$,
so $z_{0.90} = 1.282$.

Select one percentile for the two-sided significance level and one for the desired power.

Applying the approximation to the Gel X experiment

PLANNING ASSUMPTIONS

SPECIFIED TRUE EFFECT

δ = 10 percentage points

ANTICIPATED WITHIN-GROUP SD

σ = 13 percentage points

Use an SD from comparable previous studies or relevant pilot data.

DESIGN AND ERROR RATES

Equal allocation
Two-sided α = 0.05
Power: 80%

NORMAL APPROXIMATION

Substitute the Gel X planning assumptions

$$n \approx \frac{2(13)^2(1.960 + 0.842)^2}{10^2} = 26.5$$

26.5 → round upward

Normal approximation → 27 per group
The Normal approximation is slightly optimistic for a small sample.

Exact t-test calculation: n = 27.5, then round upward

28 **animals per group**
56 animals in total

Exercise: plan a new housing experiment

Researchers plan another comparison of standard and enriched recovery housing.

Target reduction: 10 ng/mL · Anticipated within-group SD: 20 ng/mL
Equal allocation · Two-sided α = 0.05 · Desired power: 90%

1. Calculate the required number of animals per group and in total using the Normal approximation.

2. How many animals would be required if the desired power were 80%?

$$n \approx \frac{2\sigma^2\left(z_{1-\alpha/2} + z_{1-\beta}\right)^2}{\delta^2}$$

p	z_p
0.80	0.842
0.85	1.036
0.90	1.282
0.95	1.645
0.975	1.960
0.99	2.326
0.995	2.576

The t-test formula is one example of a general planning principle

For any study, choose n so that the planned analysis can detect a specified scientifically relevant effect with the desired power.

1

Define the outcome and effect of interest

What comparison or population parameter should the experiment estimate?

2

Specify the situation under which power is calculated

State the effect under H_1 and the required population assumptions—for example, an SD for a continuous outcome or group probabilities for a binary outcome.

3

Match the calculation to the design and planned analysis

Comparisons of means, proportions, several treatment groups, blocked designs and repeated measurements require different calculations.

Different studies require different formulas—or sometimes simulation—but the underlying planning logic is the same.

Exercise: planning an experiment with a binary outcome

Researchers compare two recovery protocols. Each animal is assessed for a postoperative complication: yes or no.

PLANNING ASSUMPTIONS

Complication risk: standard 40% → new protocol 20%
Independent animals · Random allocation · Equal group sizes
Two-sided $\alpha = 0.05$ · Desired power: 80%

1. Use the calculator to determine the required number of animals per group and in total.

2. Repeat for a smaller reduction: 40% to 30%. Explain the change in sample size.

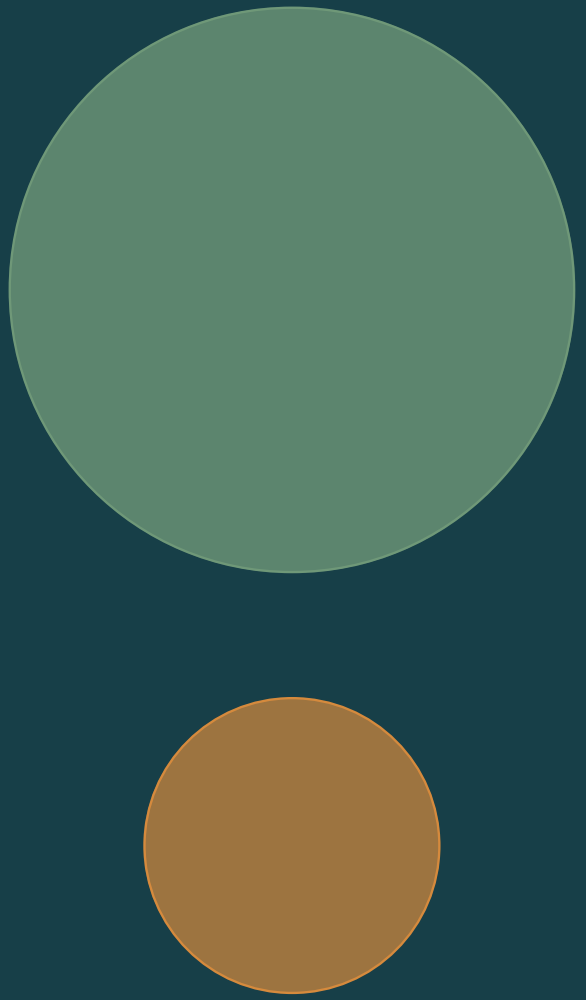
TWO-PROPORTION CALCULATOR

Open UBC calculator

Select sample size and a two-sided test. Enter proportions as decimals: 0.40 and 0.20.

PART III

Handling nuisance variables



Can comparing similar animals improve precision?

Observed group difference = true treatment effect
+ contributions from other observed and unobserved factors
that differ between the groups

RANDOMIZATION

Other contributions are zero on average.

Chance imbalance still adds variation between experiments.

IMPROVING PRECISION

Sex predicts wound closure in our example.

What if we compare treatments within each sex, then combine the comparisons?

Use information about important animal characteristics in the treatment comparison.

Stratification: compare within strata, then combine

1 **Form strata using an important animal characteristic**
Here, the strata are females and males. Both treatments must be represented within each sex.

2 **Estimate the treatment difference within each stratum**
Compare Gel X with vehicle among females, then among males. The shared sex effect cancels within each comparison.

3 **Combine the differences for the population of interest**
For our population with equal numbers of females and males:
overall effect estimate = ½ × female difference + ½ × male difference.

How does this estimate compare with the difference between the overall group means?

Exercise: calculate two treatment-effect estimates

Illustrative results for 40 individually housed mice after simple randomization to standard or enriched recovery housing. Age composition differs by chance.

Target population: equal proportions of younger and older animals. Corticosterone means in ng/mL.

OVERALL RESULTS

Housing condition	n	Mean
Standard	20	55
Enriched	20	35

RESULTS WITHIN AGE GROUPS

Age group	Standard n	Standard mean	Enriched n	Enriched mean
Younger	5	40	15	30
Older	15	60	5	50

1. Calculate the overall difference: enriched − standard.

2. Calculate the treatment difference within each age group.

3. Combine the age-specific differences using equal weights.

4. Explain why the overall and age-stratified estimates differ.

Does a stratified analysis improve precision?

Compare two analyses of the same simply randomized experiment in each repetition.

THE SIMULATED POPULATION

Gel X adds 10 percentage points in both sexes.

Females have 12 points greater expected closure.

Each experiment has 12 females and 12 males, plus other animal variation.

REPEAT 10,000 EXPERIMENTS

Randomly allocate 12 mice to each treatment.

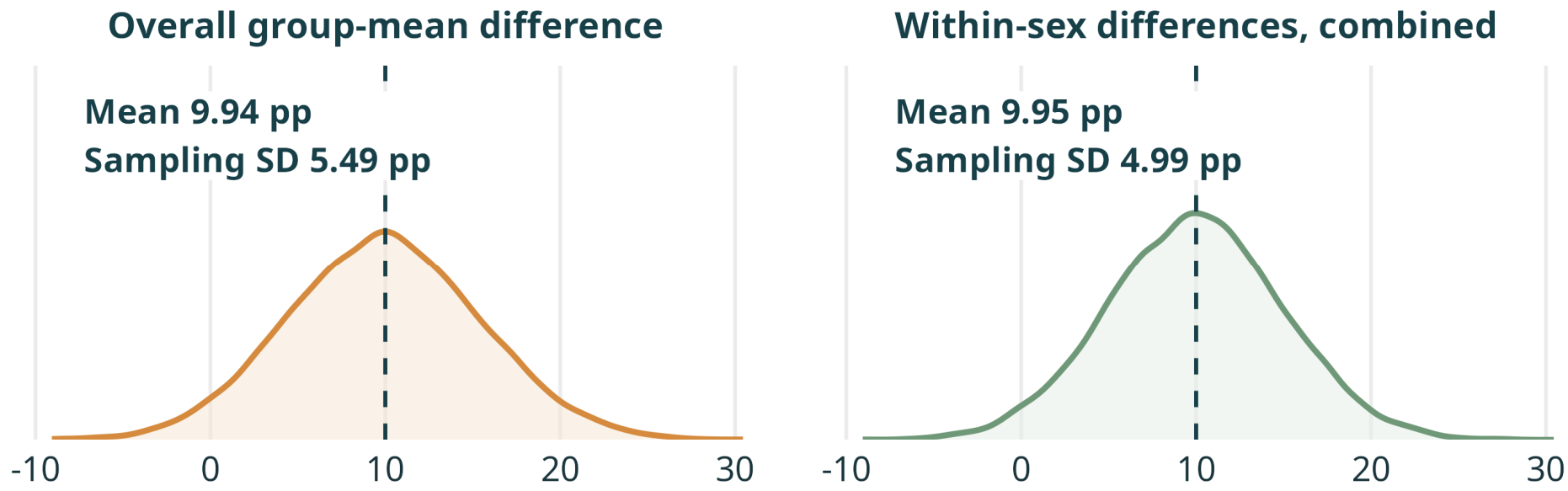
Unadjusted: subtract the two group means.

Stratified: compare within each sex, then average the two differences.

Keep the animals, allocation and outcomes the same; change how treatments are compared.

Within-stratum comparisons reduce variation between experiments

Same simple randomization in both panels · 10,000 experiments



Dashed line: true effect = +10 pp · equal female and male weights

Both estimates are unbiased here; comparing within sex improves precision.

Stratified randomization strengthens the within-stratum comparisons

Randomize six mice to each treatment within each sex.

WHY BALANCE THE ALLOCATION?

Both treatments are represented in every stratum.

For a fixed stratum size and equal outcome variances, a 1:1 split minimizes the variance of its treatment difference.

ALLOCATION WITHIN EACH STRATUM

SEX	VEHICLE	GEL X
Female	**6**	**6**
Male	**6**	**6**

Simulation: sampling SD 4.99 → 4.86 pp when within-sex comparisons also use balanced allocation.

Stratified analysis improves precision; balanced allocation can improve it further.

With exact balance, the two point estimates coincide

1 **Both treatment groups have the same sex composition**
Each group contains six females and six males. The overall mean difference already equals ½ × female difference + ½ × male difference.

2 **The point estimate is unchanged by including sex**
In one illustrative stratified allocation: +9.2 percentage points with either analysis. This is a different allocation from the original Gel X experiment.

3 **The standard error must still account for the strata**
Across repetitions: actual sampling SD = 4.9 pp.
Mean SE: treatment only = 5.5 pp; treatment + sex = 4.8 pp.

The usual unadjusted SE includes sex variation that balance removes from the comparison.

Exercise: compare the estimates with balanced allocation

Repeat the housing example with 10 younger and 10 older mice randomly allocated to each condition. Keep the age-specific means unchanged.

Target population: equal proportions of younger and older animals. Corticosterone means in ng/mL.

OVERALL RESULTS

Housing condition	n	Mean
Standard	20	50
Enriched	20	40

RESULTS WITHIN AGE GROUPS

Age group	Standard n	Standard mean	Enriched n	Enriched mean
Younger	10	40	10	30
Older	10	60	10	50

1. Calculate the overall difference: enriched − standard.

2. Calculate the treatment difference within each age group.

3. Combine the age-specific differences using equal weights.

4. Compare the two estimates. What changed from the earlier exercise?

Can we now ignore age when calculating the standard error?

Plan the comparisons, allocation and analysis together

1 **Choose an important prognostic characteristic**
Use a characteristic known before treatment that predicts the outcome. Define the population effect and the weights for combining strata.

2 **Compare within strata, then combine**
This can improve precision after simple randomization. A treatment-plus-sex regression can estimate the common effect assumed in our example.

3 **Balance the allocation within strata when useful**
Stratified randomization ensures informative within-stratum comparisons. Analyse using the strata and calculate uncertainty accordingly.

Keep relevant animal diversity while accounting for its contribution to outcome variation.

From animal characteristics to experimental context

The same principle: compare treatments within relevant groups, then combine.

STRATA BASED ON THE ANIMALS

Sex or age group.

Compare treatments within each stratum, then combine.

Randomize within strata to balance the comparisons.

BLOCKS BASED ON SHARED CONDITIONS

Session, day or batch.

Compare treatments within each block, then combine.

Randomize within blocks to balance the comparisons.

These labels overlap: animal characteristics can also define blocks.

Stratified randomization and randomized blocking use the same design principle.

The Gel X experiment is conducted across six sessions

EXPERIMENTAL SETTING

- Randomized experiment involving 24 adult mice
- Treatments: Gel X and vehicle
- Outcome: percentage wound closure after seven days
- A session is one shared run under approximately the same operational conditions
- At most four mice can be processed per session, so six sessions are required

DESIGN QUESTION

How should Gel X and vehicle be allocated across the six sessions?



Randomized blocking puts both treatments in every session

Within each session, randomize two mice to vehicle and two to Gel X—and randomize procedural order.

TREATMENT TRACKS SESSION

SESSIONS 1–3	4 × VEHICLE
SESSIONS 4–6	4 × GEL X

Treatment and session are confounded.

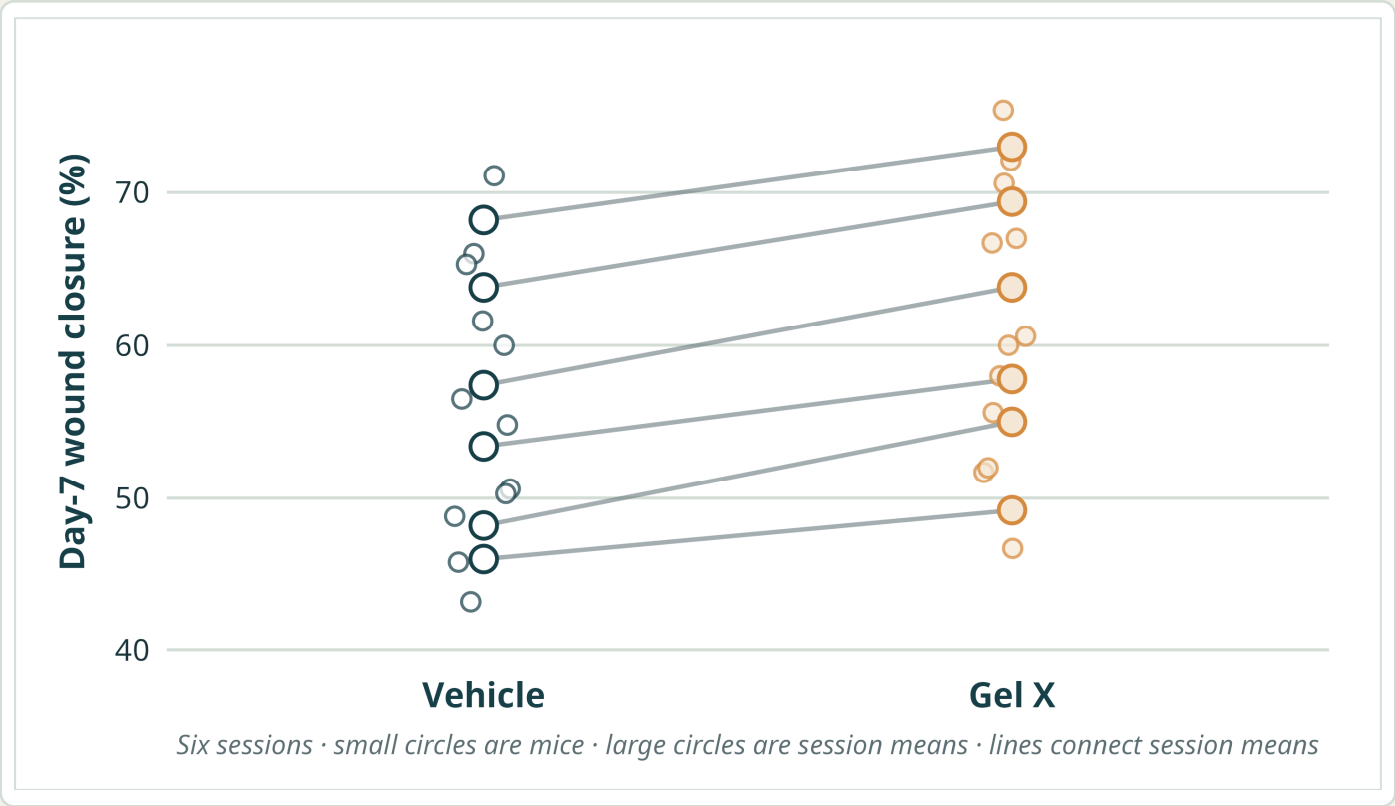
RANDOMIZE WITHIN EACH SESSION

SESSION 1	VEHICLE	GEL X	VEHICLE	GEL X
SESSION 2	GEL X	VEHICLE	GEL X	VEHICLE
⋮				
SESSION 6	VEHICLE	GEL X	GEL X	VEHICLE

Every session supplies one comparison under shared conditions.

Randomization within sessions prevents treatment from becoming confounded with session.

Blocking makes Gel X–vehicle differences visible within sessions



READ WITHIN A SESSION

1 EACH SESSION CAN DIFFER

Their shared conditions can shift the outcome level.

2 COMPARE WITHIN A LINE

Gel X and vehicle share the same session context.

3 COMBINE THE EVIDENCE

Combine the six within-session comparisons.

Blocking changes the comparison from differences between all mice to differences within shared experimental context.

The same effect estimate can have different uncertainty

Because treatment is balanced within every session, both analyses estimate the same +5.2-point effect.

ANALYSIS IGNORES SESSIONS

INDEPENDENT-SAMPLES ANALYSIS

COMPARISON	All 12 vehicle mice are compared with all 12 Gel X mice.
SESSION VARIATION	Session-to-session shifts remain mixed into residual variation.

Estimate +5.2 pp · SE 3.6 · p = 0.167

ANALYSIS USES SESSIONS

SESSION-ADJUSTED ANALYSIS

COMPARISON	Compare Gel X with vehicle within each session.
SESSION VARIATION	Shared session shifts are separated from the treatment comparison.

Estimate +5.2 pp · SE 1.3 · p = 0.001

The session-adjusted analysis matches the design and separates session shifts from the treatment comparison.

Randomized blocking: design and analysis work together



Other safeguards still matter: standardize procedures, blind people and assessors, and record the experimental context.

Blocking protects interpretation and can improve precision when blocks predict the outcome.

When should we use stratification or blocking?

1 **Simple randomization remains a valid design**
It protects against systematic allocation bias, although chance imbalance can reduce precision.

2 **Use important animal characteristics in the comparison**
Compare within prognostic strata or adjust for them in the analysis. Stratified randomization can strengthen these comparisons by ensuring balance.

3 **Block when experimental conditions vary**
If experiments span different sessions, days or batches, representing both treatments within each helps keep the comparison fair.

4 **Choose a manageable design**
We do not need to balance every recorded characteristic or create blocks for every possible source of variation.

Choose prognostic variables for the analysis and useful restrictions on allocation before the experiment.

Sample-size planning should reflect the design and analysis

1 **Our starting point in Section 2**
We planned a simple randomized experiment with equal allocation and an independent-samples t-test: a common starting point for sample-size calculations.

2 **Accounting for stratification or blocking**
A calculation tailored to the design and analysis needs assumptions about how much variation they will remove and the expected precision gain.

3 **Avoid optimistic assumptions**
Overestimating the precision gain can lead to choosing too few animals and achieving less power than intended.

4 **A practical starting point**
The simpler two-group calculation can be conservative because it does not credit the expected precision gain. Check this for the planned design.

Only reduce the planned sample size for an expected precision gain when that gain is supported by credible information.

Beyond parallel-group designs

The scientific comparison, experimental unit and structure of the observations determine which design is appropriate.

1

Define what is randomized

Treatment may be assigned to individual animals, groups of animals or treatment sequences.

2

Determine where the informative comparisons occur

Comparisons may be between independent groups or within animals, matched pairs, blocks or shared contexts.

3

Plan the design and analysis together

The analysis must respect the randomization unit, blocking, repeated observations and dependencies between measurements.

Parallel groups · paired or crossover · randomized blocks · cluster-randomized · factorial designs

Different designs organize the evidence differently; the analysis and sample-size calculation must follow.

THE END

Questions



